

MINISTRY OF EDUCATION AND SCIENCE OF THE RUSSIAN FEDERATION

Federal State Autonomous Educational
Institution of Higher Education "Southern Federal University"
(SOUTHERN FEDERAL UNIVERSITY)

COURSE SYLLABUS FOR
Information Technology Security
(Безопасность информационных технологий)

Подписано электронной подписью:

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I. GOALS AND OBJECTIVES OF MASTERING THE DISCIPLINE

Objectives of mastering the discipline:

- Formation of students' basic knowledge on the problems of information technology security, methods and means of ensuring information security, including the use of artificial intelligence
- Obtaining by students of primary skills in working with information security tools

Objectives of mastering the discipline are to provide the basic knowledge about:

- Possible threats to data security and vulnerabilities of information systems
- Classification of methods and means of information security
- Standardization in the field of information security
- Methods and means of cryptography and ways of their application
- Methods and means of authentication of subjects, objects and processes
- Methods and mechanisms of authorization of subjects in information systems
- Methods and means of ensuring the security of operating systems
- Malware detection and neutralization tools
- Methods and means of information protection in computer networks
- Technical means of information security
- Organizational and legal support of information security
- Application of artificial intelligence in information security

II. THE PLACE OF THE DISCIPLINE IN THE STRUCTURE OF THE EDUCATIONAL PROGRAM

The discipline belongs to the compulsory part of the educational program and is included in the .

Name disciplines (modules), practices	Required knowledge, skills, and abilities
Discrete Mathematics	Knowledge: – Binary Number System and Boolean Algebra – Combinatorics – Probability Theory and Mathematical Statistics – Mathematical Logic – Algebra and Number Theory – Computational Mathematics – Discrete Mathematics
Algorithmization and programming	Knowledge: – C or Python programming languages – Data Structures and Data Processing Algorithms Skills: – Work in a text editor with syntax highlighting or in a development environment for editing the source code of programs in the C or Python programming language – Compiling the source code of programs (written in the C programming language) into executable binaries – Debugging programs with debuggers
Operating Systems	Knowledge: – Windows and Linux operating system architectures – Resource allocation and allocation principles in multitasking multi-user operating systems

Name disciplines (modules), practices	Required knowledge, skills, and abilities
	Skills: – Installing, running, and configuring Windows and Linux operating systems, including network services – Installing, configuring, and running software on Windows and Linux operating systems

Knowledge, skills and abilities formed by this discipline will be required when mastering the following elements of the educational program:

Production, pre-graduation practice;

Preparation for the defense and defense of the final qualification work.

III. REQUIREMENTS FOR THE RESULTS OF MASTERING THE DISCIPLINE

The development of the discipline is aimed at the formation of the following competencies in accordance with the educational standard and educational program:

List of planned learning outcomes in the discipline correlated with indicators of competence achievement

Competence	Indicators of Competency Achievement	Learning outcomes
CPC-2. Able to solve the problems of professional activity using modern information and communication technologies, artificial intelligence methods and technologies, hardware and software, including domestic production, and taking into account the basic requirements of information security and professional ethics	CPC-2.1. Selects modern information technologies and software, including domestic ones, to solve the problems of professional activity	Knowledge: - Possible threats to data security and vulnerabilities of information systems - Classification of methods and means of information protection - Standardization in the field of information security
	CPC-2.3. Solves standard tasks of professional activity on the basis of information and bibliographic culture with the use of information and communication technologies	Knowledge: - Organizational and legal support of information security
		Skills: - Development of basic requirements for ensuring information security using modern information and communication technologies
	CPC-2.4. Takes into account the basic requirements of information security when solving standard tasks of professional activity	Knowledge: - Methods and means of cryptography and ways of their application - Methods and means of authentication of subjects, objects and processes - Methods and mechanisms of authorization of subjects in information systems - Methods and tools for ensuring the security of operating systems - Malware detection and neutralization tools; Methods and Means of Information Protection in Computer Networks - Technical means of information security
		Skills: - Solve the problems of ensuring information security using modern information and communication technologies

Competence	Indicators of Competency Achievement	Learning outcomes
CPC-5. Capable of installing software and hardware for information and automated systems	CPC-5.1. Installs software based on modern standards of information interaction of systems	Knowledge: - Methods and tools for ensuring the security of operating systems - Malware detection and neutralization tools
		Skills: - Installation and configuration of a software firewall
	CPC-5.2. Performs configuration of software for information and automated systems	Knowledge: - Technical means of information security
		Skills: - Work with technical means of information security
PC-13. AI S-1 Able to manage risks in the development of AI systems, build AI security management in the company with AI ethics in mind	PC-13.1. AI S -1.1 Identifies and models threats throughout the life cycle of AI systems, assesses and prioritizes risks	Knowledge: – knows about the threats and risks of using artificial intelligence – Methods for Risk Detection and Mitigation
	PC-13.2 AI S -1.2 Ensures compliance with regulatory requirements and trusted/ethical AI principles	Knowledge: – Principles of Legal Regulation of Artificial Intelligence – Ethics of Artificial Intelligence

IV. CONTENT AND STRUCTURE OF THE DISCIPLINE

The complexity of the discipline is 5 credits, 180 hours
1 credit, 36 hours per exam

Form of intermediate certification: exam

4.1. The content of the discipline, structured by topics (educational meetings) for the 5th semester of full-time study

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
Module 1.										
1	Basic concepts and definitions. Threats and vulnerabilities of information systems. Data protection criteria and standards. Goals and objectives of mastering the discipline. Introduction to Information Security. Basic concepts and definitions. Classification of sources of information security threats. Classification of threats to information security of an information system. Ensuring the security of the infrastructure of the information system.	2	—	—	—	2	Informational lecture	—	—	—

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
2	Basic concepts and definitions. Threats and vulnerabilities of information systems. Data protection criteria and standards. Methods for assessing threats to the security of an information system. Classification of intruders. Channels of penetration into the information system and channels of information leakage. Approaches to ensuring information security of information systems. Data protection criteria and standards.	2	–	–	–	2	Informational lecture	–	–	–
3	Authentication methods and tools. Basic concepts. Password authentication. One-time passwords. Functional authentication methods. Personal means of authentication. Biometric means of authentication. Authentication based on information associated with the subject. Multi-channel authentication.	2	–	–	–	2	Informational lecture	–	–	–

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
4	Methods and means of authorization. Access control. Security policies and models. Discretionary safety model. Mandatory security model. Role-based model of access control. A model of information flow security. Sandbox model. Thematic access model.	2	—	—	—	2	Informational lecture	—	—	—

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
5	Technical protection of information. Technical channels of information leakage. Use of technical channels of information leakage for technical intelligence. Classification of technical channels of information leakage. Technical channels of information leakage in the main technical means and systems. Electromagnetic leakage channels in the main technical means and systems. Electrical channels of information leakage in the main technical means and systems. Parametric channels of information leakage in the main technical means and systems. Technical channels of information leakage during data transmission. Mobile radio communication. Radio relay and space communication systems. Electrical channels of information leakage in wire communication lines. Interception of information from wire electrical communication lines. Interception of information from telephone communication lines. Interception of information from radiotelephone communication lines.	2	–	–	–	2	Informational lecture	–	–	–

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
6	Technical protection of information. Technical channels of speech information leakage. Speech information. Acoustic channels of speech information leakage. Vibroacoustic channels of speech information leakage. Acoustoelectric channels of speech information leakage. Acousto-fiber-optic channels for speech information leakage. Optoelectronic leakage channels speech information. Parametric channels of speech information leakage. Technical channels of leakage of visual information. Covert surveillance of objects. Optical-mechanical devices. Thermal imagers and night vision devices. Covert photography of objects of observation. Covert photography of documents. Covert video surveillance. Material channels of information leakage. Integration of information leakage channels.	2	—	—	—	2	Informational lecture	—	—	—

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
7	Organizational and legal support of information security. Classification of information security methods. Classification and types of information resources. Restricted information. State secret. Confidential Information. Trade secret. Confidentiality stamps. Legal basis of the licensing and certification system. Licensing of activities related to state secrets and information protection. Certification of information security tools. Certification of activities for the technical protection of confidential information. Ethics of Artificial Intelligence. Legal regulation of artificial intelligence.	2	–	–	–	2	Informational lecture	–	–	–

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
8	Artificial Intelligence in Information Security. Problems of information security of artificial intelligence. Application of artificial intelligence in information security. Classification of information security systems using artificial intelligence. Classification of computer attacks on artificial intelligence. Methods of countering attacks on artificial intelligence. Products for information protection based on artificial intelligence. Directions for the development of artificial intelligence in information security.	2	–	–	–	2	Informational lecture	–	–	–
9	Artificial Intelligence in Information Security. Federated learning. Differential confidentiality. Homomorphic encryption. Software products to protect the confidentiality and integrity of artificial intelligence data. Data protection from infection. Anomaly recognition methods. Test No1.	2	–	–	–	2	Informational lecture	Test No1 (testing)	–	10

Module 2.

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
10	Cryptographic methods of information protection and cryptanalysis. Symmetric and asymmetric algorithms (AES, GOST, Grasshopper, RSA). Hashing, EDS, PKI. AI threats: accelerated key guessing, ciphertext analysis. Requirements for durability and certification.	2	–	–	–	2	Informational lecture	–	–	–
11	Cryptographic methods of information protection and cryptanalysis. Typical implementation errors: weak RNG, replay IV, poor key management. Cryptanalysis methods. The role of AI in automating attacks on cryptosystems. FSB requirements for domestic algorithms. Post-quantum cryptography.	2	–	–	–	2	Informational lecture	–	–	–

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
12	Cryptographic methods of information protection and cryptanalysis. Symmetric and asymmetric algorithms (AES, GOST, Grasshopper, RSA). Hashing, EDS, PKI. AI threats: accelerated key guessing, ciphertext analysis. Requirements for strength and certification. Typical implementation errors: weak RNG, replay IV, poor key management. Cryptanalysis methods. The role of AI in automating attacks on cryptosystems. FSB requirements for domestic algorithms. Post-quantum cryptography.	–	2	–	–	2	Seminar	–	–	–
13	Cryptographic methods of information protection and cryptanalysis. TLS, IPsec, PGP protocols. Security in mobile and IoT devices. Use of AI to monitor PKI integrity and detect anomalies in cryptosystems. Incident analysis: Heartbleed, Logjam. Update and support of cryptographic libraries.	2	–	–	–	2	Informational lecture	–	–	–

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
14	Cryptographic methods of information protection and cryptanalysis. TLS, IPsec, PGP protocols. Security in mobile and IoT devices. Use of AI to monitor PKI integrity and detect anomalies in cryptosystems. Incident analysis: Heartbleed, Logjam. Update and support of cryptographic libraries.		2	–	–	2		Seminar	–	–
15	Malicious software. Classification: viruses, worms, trojans, rootkits, ransomware. Polymorphism and metamorphism. Use of AI by cybercriminals: phishing generation, deepfake, EDR bypass. Use of AI in next-generation antivirus systems.	2	–	–	–	2	Informational lecture	–	–	–
16	Malicious software. Classification: viruses, worms, trojans, rootkits, ransomware. Polymorphism and metamorphism. Use of AI by cybercriminals: phishing generation, deepfake, EDR bypass. Use of AI in next-generation antivirus systems.	–	2	–	–	2	Seminar	–	–	–

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
17	Malicious software. Targeted attacks (APTs), multi-stage chains of compromise. AI for analyzing software behavior in the sandbox. Examples: Stuxnet, WannaCry, NotPetya. Countermeasures: backup, patching, cyber hygiene.	2	—	—	—	2	Informational lecture	—	—	—
18	Malicious software. Targeted attacks (APTs), multi-stage chains of compromise. AI for analyzing software behavior in the sandbox. Examples: Stuxnet, WannaCry, NotPetya. Countermeasures: backup, patching, cyber hygiene.	—	2	—	—	2	Seminar	—	—	—
19	Security of operating systems. Security architecture: privilege levels, process isolation. Mechanisms: SELinux, AppArmor, UAC, SIP. Vulnerabilities: buffer overflow, race conditions. Use of AI to monitor system calls and detect rootkits.	2	—	—	—	2	Informational lecture	—	—	—

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
20	Security of operating systems. Security architecture: privilege levels, process isolation. Mechanisms: SELinux, AppArmor, UAC, SIP. Vulnerabilities: buffer overflow, race conditions. Use of AI to monitor system calls and detect rootkits.	–	2	–	–	2	Seminar	–	–	–
21	Security of operating systems. Kernel protection, TPM, Secure Boot. Virtualization and containerization as isolation tools. AI for automatic configuration of security policies and OS integrity analysis. Rating standards: Common Criteria, EAL.	2	–	–	–	2	Informational lecture	–	–	–
22	Security of operating systems. Kernel protection, TPM, Secure Boot. Virtualization and containerization as isolation tools. AI for automatic configuration of security policies and OS integrity analysis. Rating standards: Common Criteria, EAL.	–	2	–	–	2	Seminar	–	–	–

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
23	Protection of information in computer networks. Network attacks: MITM, ARP-spoofing, DDoS. Security tools: firewall, IDS/IPS, segmentation. Protocols: IPsec, TLS, WPA3. AI for encrypted traffic analysis, anomaly detection, and automatic network segmentation.	2	–	–	–	2	Informational lecture	–	–	–
24	Protection of information in computer networks. Network attacks: MITM, ARP-spoofing, DDoS. Security tools: firewall, IDS/IPS, segmentation. Protocols: IPsec, TLS, WPA3. AI for encrypted traffic analysis, anomaly detection, and automatic network segmentation.	–	2	–	–	2	Seminar	–	–	–
25	Information protection in computer networks. Zero Trust architectures, SDP, secure remote access. AI in cloud and hybrid information security. Threat prediction and response automation. Standards: NIST CSF, CIS Controls.	2	–	–	–	2	Informational lecture	Test No2 (testing)	–	10

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
26	Information protection in computer networks. Zero Trust architectures, SDP, secure remote access. AI in cloud and hybrid information security. Threat prediction and response automation. Standards: NIST CSF, CIS Controls.	—	2	—	—	2	Seminar	—	—	—
27	Analysis of spurious electromagnetic emissions using AI for anomaly detection Using machine learning to classify EMR and identify hidden transmitters based on spectrograms.	—	—	2	—	2	Reproductive laboratory work	Laboratory Work No 1	<i>Preparation and submission of a report on laboratory work</i>	5
28	Protection against information leakage via a vibroacoustic channel using AI filtering Simulation of vibroacoustic leaks and the use of neural network filters to suppress informative components.	—	—	2	—	2	Reproductive laboratory work	Laboratory Work No 2	<i>Preparation and submission of a report on laboratory work</i>	5
29	Development and analysis of an AI-controlled speech signal scrambler Study of traditional and adaptive (AI-based) methods of speech masking; Assessment of resistance to automatic recovery using speech-to-text models.	—	—	2	—	2	Reproductive laboratory work	Laboratory Work No 3	<i>Preparation and submission of a report on laboratory work</i>	5

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)					Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Practical	Laboratory	Consulting					
30	Detection of inhomogeneities in cable lines using AI analysis of reflectograms Use of convolutional neural networks (CNN) or time-series methods to automatically identify connection points of listening devices.	–	–	2	–	2	Reproductive laboratory work	Laboratory Work No 4	<i>Preparation and submission of a report on laboratory work</i>	5
31	Password recovery using AI-accelerated brute-force and leak analysis Using recurrent neural networks (RNNs) or transformers to generate probable passwords based on leak statistics (for example, from HaveIBeenPwned).	–	–	2	–	2	Reproductive laboratory work	Laboratory Work No 5	<i>Preparation and submission of a report on laboratory work</i>	5
32	Analysis of the Resilience of the Grasshopper Cryptoalgorithm to AI Attacks Exploring the possibility of using machine learning to find correlations between plaintext and ciphertext.	–	–	2	–	2	Reproductive laboratory work	Laboratory Work No 6	<i>Preparation and submission of a report on laboratory work</i>	5
33	Malware detection using ML models based on static and dynamic features Training a classifier (for example, Random Forest or neural networks) on a dataset of malicious/legitimate executable files.	–	–	4	–	4	Reproductive laboratory work	Laboratory Work No 7	<i>Preparation and submission of a report on laboratory work</i>	5

No p/n	Topic of the lesson and issues under consideration	Types of educational work and their labor intensity, hours (including the use of online courses)				Format of the lesson	Name of the Valuation Tool	Type of control	Sum of points per lesson	
		Contact work								Independe nt work
		Lectures	Practical	Laboratory	Consulting					
34	AI-powered network traffic analysis to detect attacks (including encrypted traffic) Classification of traffic (normal / DDoS / MITM) based on metadata and temporal patterns without decryption.	–	–	4	–	4	Reproductive laboratory work	Laboratory Work No 8	<i>Preparation and submission of a report on laboratory work</i>	5
Intermediate certification <i>(for disciplines with an exam)</i>		–	–	–	–	36	–	Exam Questions and Tickets for Semester 5	<i>Exam</i>	40
Total Hours		36	16	20	–	180	–	–	–	40

4.2. Plan of extracurricular independent work

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
Module 1.						

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
1	Basic concepts and definitions. Threats and vulnerabilities of information systems. Data protection criteria and standards. Goals and objectives of mastering the discipline. Introduction to Information Security. Basic concepts and definitions. Classification of sources of information security threats. Classification of threats to information security of an information system. Ensuring the security of the infrastructure of the information system.	5	1	Study and revision of lecture material	2	[1], [2]
2	Basic concepts and definitions. Threats and vulnerabilities of information systems. Data protection criteria and standards. Methods for assessing threats to the security of an information system. Classification of intruders. Channels of penetration into the information system and channels of information leakage. Approaches to ensuring information security of information systems. Data protection criteria and standards.	5	2	Study and revision of lecture material	2	[1], [2], [9]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
3	Authentication methods and tools. Basic concepts. Password authentication. One-time passwords. Functional authentication methods. Personal means of authentication. Biometric means of authentication. Authentication based on information associated with the subject. Multi-channel authentication.	5	3	Study and revision of lecture material	2	[1] – [3], [9]
4	Methods and means of authorization. Access control. Security policies and models. Discretionary safety model. Mandatory security model. Role-based model of access control. A model of information flow security. Sandbox model. Thematic access model.	5	4	Study and revision of lecture material	2	[1] – [3], [9]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
5	Technical protection of information. Technical channels of information leakage. Use of technical channels of information leakage for technical intelligence. Classification of technical channels of information leakage. Technical channels of information leakage in the main technical means and systems. Electromagnetic leakage channels in the main technical means and systems. Electrical channels of information leakage in the main technical means and systems. Parametric channels of information leakage in the main technical means and systems. Technical channels of information leakage during data transmission. Mobile radio communication. Radio relay and space communication systems. Electrical channels of information leakage in wire communication lines. Interception of information from wire electrical communication lines. Interception of information from telephone communication lines. Interception of information from radiotelephone communication lines. Channels for information leakage from fiber-optic communication lines.	5	5	Study and revision of lecture material	2	[4]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
6	Technical protection of information. Technical channels of speech information leakage. Speech information. Acoustic channels of speech information leakage. Vibroacoustic channels of speech information leakage. Acoustoelectric channels of speech information leakage. Acousto-fiber-optic channels for speech information leakage. Optoelectronic leakage channels speech information. Parametric channels of speech information leakage. Technical channels of leakage of visual information. Covert surveillance of objects. Optical-mechanical devices. Thermal imagers and night vision devices. Covert photography of objects of observation. Covert photography of documents. Covert video surveillance. Material channels of information leakage. Integration of information leakage channels.	5	6	Study and revision of lecture material	2	[4]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
7	Organizational and legal support of information security. Classification of information security methods. Classification and types of information resources. Restricted information. State secret. Confidential Information. Trade secret. Confidentiality stamps. Legal basis of the licensing and certification system. Licensing of activities related to state secrets and information protection. Certification of information security tools. Certification of activities for the technical protection of confidential information. Ethics of Artificial Intelligence. Legal regulation of artificial intelligence.	5	7	Study and revision of lecture material	2	[1]–[3], [8]
8	Artificial Intelligence in Information Security. Problems of information security of artificial intelligence. Application of artificial intelligence in information security. Classification of information security systems using artificial intelligence. Classification of computer attacks on artificial intelligence. Methods of countering attacks on artificial intelligence. Products for information protection based on artificial intelligence. Directions for the development of artificial intelligence in information security.	5	8	Study and revision of lecture material	2	[1]–[3], [9]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
9	Artificial Intelligence in Information Security. Federated learning. Differential confidentiality. Homomorphic encryption. Software products to protect the confidentiality and integrity of artificial intelligence data. Data protection from infection. Anomaly recognition methods. Test No1.	5	9	Study and revision of lecture material	2	[1]–[3], [9]
Module 2.						
10	Cryptographic methods of information protection and cryptanalysis. Symmetric and asymmetric algorithms (AES, GOST, Grasshopper, RSA). Hashing, EDS, PKI. AI threats: accelerated key guessing, ciphertext analysis. Requirements for durability and certification.	5	10	Study and revision of lecture material	2	[4]
11	Cryptographic methods of information protection and cryptanalysis. Typical implementation errors: weak RNG, replay IV, poor key management. Cryptanalysis methods. The role of AI in automating attacks on cryptosystems. FSB requirements for domestic algorithms. Post-quantum cryptography.	5	11	Study and revision of lecture material	2	[4]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
12	Cryptographic methods of information protection and cryptanalysis. Symmetric and asymmetric algorithms (AES, GOST, Grasshopper, RSA). Hashing, EDS, PKI. AI threats: accelerated key guessing, ciphertext analysis. Requirements for strength and certification. Typical implementation errors: weak RNG, replay IV, poor key management. Cryptanalysis methods. The role of AI in automating attacks on cryptosystems. FSB requirements for domestic algorithms. Post-quantum cryptography.	5	11	Study and revision of the material of practical classes	2	[4]
13	Cryptographic methods of information protection and cryptanalysis. TLS, IPsec, PGP protocols. Security in mobile and IoT devices. Use of AI to monitor PKI integrity and detect anomalies in cryptosystems. Incident analysis: Heartbleed, Logjam. Update and support of cryptographic libraries.	5	12	Study and revision of lecture material	2	[4]
14	Cryptographic methods of information protection and cryptanalysis. TLS, IPsec, PGP protocols. Security in mobile and IoT devices. Use of AI to monitor PKI integrity and detect anomalies in cryptosystems. Incident analysis: Heartbleed, Logjam. Update and support of cryptographic libraries.	5	12	Study and revision of the material of practical classes	2	[4]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
15	Malicious software. Classification: viruses, worms, trojans, rootkits, ransomware. Polymorphism and metamorphism. Use of AI by cybercriminals: phishing generation, deepfake, EDR bypass. Use of AI in next-generation antivirus systems.	5	13	Study and revision of lecture material	2	[4]
16	Malicious software. Classification: viruses, worms, trojans, rootkits, ransomware. Polymorphism and metamorphism. Use of AI by cybercriminals: phishing generation, deepfake, EDR bypass. Use of AI in next-generation antivirus systems.	5	13	Study and revision of the material of practical classes	2	[4]
17	Malicious software. Targeted attacks (APTs), multi-stage chains of compromise. AI for analyzing software behavior in the sandbox. Examples: Stuxnet, WannaCry, NotPetya. Countermeasures: backup, patching, cyber hygiene.	5	14	Study and revision of lecture material	2	[4]
18	Malicious software. Targeted attacks (APTs), multi-stage chains of compromise. AI for analyzing software behavior in the sandbox. Examples: Stuxnet, WannaCry, NotPetya. Countermeasures: backup, patching, cyber hygiene.	5	14	Study and revision of the material of practical classes	2	[4]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
19	Security of operating systems. Security architecture: privilege levels, process isolation. Mechanisms: SELinux, AppArmor, UAC, SIP. Vulnerabilities: buffer overflow, race conditions. Use of AI to monitor system calls and detect rootkits.	5	15	Study and revision of lecture material	2	[1]–[3], [10]
20	Security of operating systems. Security architecture: privilege levels, process isolation. Mechanisms: SELinux, AppArmor, UAC, SIP. Vulnerabilities: buffer overflow, race conditions. Use of AI to monitor system calls and detect rootkits.	5	15	Study and revision of the material of practical classes	2	[1]–[3], [10]
21	Security of operating systems. Kernel protection, TPM, Secure Boot. Virtualization and containerization as isolation tools. AI for automatic configuration of security policies and OS integrity analysis. Rating standards: Common Criteria, EAL.	5	16	Study and revision of lecture material	2	[4]
22	Security of operating systems. Kernel protection, TPM, Secure Boot. Virtualization and containerization as isolation tools. AI for automatic configuration of security policies and OS integrity analysis. Rating standards: Common Criteria, EAL.	5	16	Study and revision of the material of practical classes	2	[4]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
23	Protection of information in computer networks. Network attacks: MITM, ARP-spoofing, DDoS. Security tools: firewall, IDS/IPS, segmentation. Protocols: IPsec, TLS, WPA3. AI for encrypted traffic analysis, anomaly detection, and automatic network segmentation.	5	17	Study and revision of lecture material	2	[1]–[3], [9]
24	Protection of information in computer networks. Network attacks: MITM, ARP-spoofing, DDoS. Security tools: firewall, IDS/IPS, segmentation. Protocols: IPsec, TLS, WPA3. AI for encrypted traffic analysis, anomaly detection, and automatic network segmentation.	5	17	Study and revision of the material of practical classes	2	[1]–[3], [9]
25	Information protection in computer networks. Zero Trust architectures, SDP, secure remote access. AI in cloud and hybrid information security. Threat prediction and response automation. Standards: NIST CSF, CIS Controls.	5	18	Study and revision of lecture material	2	[4]
26	Information protection in computer networks. Zero Trust architectures, SDP, secure remote access. AI in cloud and hybrid information security. Threat prediction and response automation. Standards: NIST CSF, CIS Controls.	5	18	Study and revision of the material of practical classes	2	[4]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
27	Analysis of spurious electromagnetic emissions using AI for anomaly detection Using machine learning to classify EMR and identify hidden transmitters based on spectrograms.	5	10	preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	2	[4], Description of laboratory work
28	Protection against information leakage via a vibroacoustic channel using AI filtering Simulation of vibroacoustic leaks and the use of neural network filters to suppress informative components.	5	11	preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	2	[5]–[7], [11], Description of laboratory work
29	Development and analysis of an AI-controlled speech signal scrambler Study of traditional and adaptive (AI-based) methods of speech masking; Assessment of resistance to automatic recovery using speech-to-text models.	5	12	preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	2	[5]–[7], [11], Description of laboratory work
30	Detection of inhomogeneities in cable lines using AI analysis of reflectograms Use of convolutional neural networks (CNN) or time-series methods to automatically identify connection points of listening devices.	5	13	preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	2	[5]–[7], [11], Description of laboratory work
31	Password recovery using AI-accelerated brute-force and leak analysis Using recurrent neural networks (RNNs) or transformers to generate probable passwords based on leak statistics (for example, from HaveIBeenPwned).	5	14	preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	2	[5]–[7], [11], Description of laboratory work

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
32	Analysis of the Resilience of the Grasshopper Cryptoalgorithm to AI Attacks Exploring the possibility of using machine learning to find correlations between plaintext and ciphertext.	5	15	preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	2	[5]–[7], [11], Description of laboratory work
33	Malware detection using ML models based on static and dynamic features Training a classifier (for example, Random Forest or neural networks) on a dataset of malicious/legitimate executable files.	5	16-17	preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	4	[5]–[7], [11], Description of laboratory work
34	AI-powered network traffic analysis to detect attacks (including encrypted traffic) Classification of traffic (normal / DDoS / MITM) based on metadata and temporal patterns without decryption.	5	17-18	preparation for laboratory work, preparation of reports on the performance of laboratory work, preparation for the defense of reports on the performance of laboratory work	4	[5]–[7], [11], Description of laboratory work
Exam preparation (for exam disciplines)					36	[1]–[8]
The total labor intensity of independent work in the discipline					108	–

4.3. Classes implemented in the form of practical training

No p/n	Topic of the lesson	Type of educational work	Format of the lesson	Location of the class	Volume in hours
1	Analysis of spurious electromagnetic emissions using AI to detect anomalies.	Laboratory work No1	Reproductive laboratory work	SFedU	4

	Using machine learning to classify EMR and identify hidden transmitters based on spectrograms.				
2	Protection against information leakage through a vibroacoustic channel using AI filtering. Simulation of vibroacoustic leaks and the use of neural network filters to suppress informative components.	Laboratory Work No2	Reproductive laboratory work	SFedU	4
3	Development and analysis of an AI-controlled speech signal scrambler. Study of traditional and adaptive (AI-based) methods of speech masking; Assessment of Resilience to Automatic Recovery Using Speech-to-Text Models	Laboratory Work No3	Reproductive laboratory work	SFedU	4
4	Detection of inhomogeneities in cable lines using AI analysis of reflectograms Use of convolutional neural networks (CNN) or time-series methods to automatically identify connection points of listening devices	Laboratory Work No4	Reproductive laboratory work	SFedU	4
5	Password recovery using AI-accelerated brute force and leak analysis. Using recurrent neural networks (RNNs) or transformers to generate probable passwords based on	Laboratory Work No5	Reproductive laboratory work	SFedU	4

	leak statistics (e.g., from HaveIBeenPwned)				
6	Analysis of the Resilience of the Grasshopper Cryptoalgorithm to AI Attacks Study of the Possibility of Using Machine Learning to Find Correlations Between Plaintext and Ciphertext	Laboratory Work No6	Reproductive laboratory work	SFedU	4
7	Malware detection using ML models based on static and dynamic features Training a classifier (for example, Random Forest or neural networks) on a dataset of malicious/legitimate executable files	Laboratory Work No7	Reproductive laboratory work	SFedU	4
8	AI-powered network traffic analysis to detect attacks (including encrypted traffic) Traffic classification (normal / DDoS / MITM) based on metadata and temporal patterns without decryption	Laboratory Work No8	Reproductive laboratory work	SFedU	4
9	Laboratory Thesis Defenses 1-8	Laboratory work No1	Reproductive laboratory work	SFedU	4
10	Laboratory Thesis Defenses 1-8	Laboratory Work No2	Reproductive laboratory work	SFedU	4
11	Laboratory Thesis Defenses 1-8	Laboratory Work No3	Reproductive laboratory work	SFedU	4
12	Laboratory Thesis Defenses 1-8	Laboratory Work No4	Reproductive laboratory work	SFedU	4
13	Laboratory Thesis Defenses 1-8	Laboratory Work No5	Reproductive laboratory work	SFedU	4

14	Laboratory Thesis Defenses 1-8	Laboratory Work No6	Reproductive laboratory work	SFedU	4
15	Laboratory Thesis Defenses 1-8	Laboratory Work No7	Reproductive laboratory work	SFedU	4
16	Laboratory Thesis Defenses 1-8	Laboratory Work No8	Reproductive laboratory work	SFedU	4
TOTAL:					64

V. EDUCATIONAL AND METHODOLOGICAL SUPPORT OF THE DISCIPLINE

5.1. References

- 1) Artemov A.V. Information Security [Elektronnyi resurs] / A.V. Artemov. – Orel: MABIV, 2014. – 257 p. <http://biblioclub.ru/index.php?page=book&id=428605>
- 2) Bryukhomitsky, Y.A. Security of Information Technologies. Part 1: Textbook. – Southern Federal University. – Rostov-on-Don – Taganrog: Publishing House of the Southern Federal University, 2020. – 207 p.
- 3) Information Security and Information Protection. – Moscow: Student Science, 2012. - 1322 p. – Rezhim dostupa URL: <http://biblioclub.ru/index.php?page=book&id=227774>.
- 4) Kreopalov V.V. Technical Means and Methods of Information Protection [Elektronnyi resurs] / V.V. Kreopalov. - Moscow: Eurasian Open Institute, 2011. – 278 p. <http://biblioclub.ru/index.php?page=book&id=90753>
- 5) Petrov A.A. Computer Security. Cryptographic Methods of Protection. – Saratov: Professional Education, 2017.– 446 p. <http://www.iprbookshop.ru/63800.html>
- 6) Shangin V.F. Zashchita komp'yuternoy informatsii [Protection of computer information]. Effective Methods and Means. – Saratov: Professional Education, 2017. – 544 p. – Rezhim dostupa URL: <http://www.iprbookshop.ru/63592.html>.
- 7) Shangin V.F. Information Security and Information Protection [Elektronnyi resurs] – Saratov: Profobrazovaniye, 2017. – 702 p. – Mode of access: URL: <http://www.iprbookshop.ru/63594.html>
- 8) Filipova I.A. Pravovoe regulirovanie iskusstvennogo intellekta [Legal regulation of artificial intelligence]: textbook, 3rd edition, updated and supplemented – Nizhny Novgorod: Nizhny Novgorod State University, 2025. – 321 p. https://law.unn.ru/wp-content/uploads/sites/18/2023/09/PRII_2025_1.pdf
- 9) Chesalin A.N. Fundamentals of Artificial Intelligence with Applications in Information Security [Elektronnyi resurs]: Uchebnoe posobie [Fundamentals of Artificial Intelligence with Applications in Information Security]: Textbook / Chesalin A.N. – M.: MIREA – Russian Technological University, 2021. – 155 c. <https://e.lanbook.com/book/182429>

5.2. Further reading

- 10) Kuryachiy G.V. Operating System Linux. Course of lectures. – Saratov: Professional Education, 2017. – 348 p. – Mode of access. <Http://www.iprbookshop.ru/63944.html>
- 11) Electronic Educational and Methodological Manual for the Discipline "Mathematical Foundations of Cryptology" / compiled by: Trunov I.L., Polikarpov S.V. – Taganrog, 2015. – 30 p. <http://open-edu.sfedu.ru/node/2755>

5.3. List of Internet resources

- Official website of FSTEC of Russia <https://fstec.ru/>
- Knowledge base of known vulnerabilities in the field of information security [Cve.mitre.org](https://cve.mitre.org/)
- Official website of the US Institute of Standards and Technology [Www.nist.gov](https://www.nist.gov)
- Information security portal with daily news from the field of information security [Www.securitylab.ru](https://www.securitylab.ru)
- Information security portal with daily news from the field of information security [Www.bugtraq.ru](https://www.bugtraq.ru)
- Official website of the Russian developer of antivirus programs and services [Www.kaspersky.ru](https://www.kaspersky.ru)
- Official website of the Russian developer of antivirus programs and services [Www.drweb.com](https://www.drweb.com)
- Portal dedicated to the study of Exelab.ru programs
- A website for a special Linux distribution designed for penetration testing [Www.kali.org](https://www.kali.org)
- [Www.metasploit.com](https://www.metasploit.com) Exploit Development and Research Tools website

- Official website of the Russian manufacturer of hardware keys and smart cards (RuToken) [Www.rutoken.ru](http://www.rutoken.ru)
- Official site of the Russian manufacturer of hardware keys and smart cards (Aladdin R.D.) [Www.aladdin-rd.ru](http://www.aladdin-rd.ru)
- The largest open source software resource [Www.github.com](http://www.github.com)
- Official website of the Eclipse [Www.eclipse.org](http://www.eclipse.org) programming environment
- Galatenko V.A. Standards of Information Security: Textbook [Elektronnyi resurs]. – INTUITION. [Http://www.intuit.ru/studies/courses/30/30/info](http://www.intuit.ru/studies/courses/30/30/info)
- Belyaev A.V. Metody i sredstva zashchite informatsii (kurs lektiv) [Methods and means of information protection (course of lectures)]. – URL access mode <http://www.citforum.ru/internet/infsecure/index.shtml>
- Artificial Intelligence of the Russian Federation. Regular Season [Https://ai.gov.ru/ai/regulatory/](https://ai.gov.ru/ai/regulatory/)
- Stepik. Ethics and Legal Problems of Artificial Intelligence [Https://stepik.org/course/118318/](https://stepik.org/course/118318/)
- Stepik. Trusted Artificial Intelligence [Https://stepik.org/course/125906/](https://stepik.org/course/125906/)
- Official website of the Technical Committee 26 of Russia (on cryptography) [Www.tk26.ru](http://www.tk26.ru)

VI. MATERIAL AND TECHNICAL SUPPORT OF THE DISCIPLINE

The following premises, equipment and software are used in the implementation of the discipline:

No p/n	Type of classroom lesson	Recommended audience	
		Type	Equipment
1	Lectures	Lecture Hall	- Interactive whiteboard 1 pc. - Teacher's computer 1 pc. - Microsoft Windows, Microsoft Office PowerPoint
2	Laboratory Classes	Laboratory in the field of software and hardware for ensuring information security	- Test bench for testing of advanced product forms of lines 1 pc. - Stand for acoustic and vibroacoustic measurements and active protection against data leakage via a vibroacoustic channel 1 pc. - Radio bookmark search stand 1 pc. - Stand for evaluation of compromising materials from radio-electronic devices 1 pc. - Piranha Device st03 1 pcs. - Lux non-linear locator 1 pc. - High-frequency generator G4-11 61 pcs. - High-frequency generator G4-10 91 pcs. - Scanning receiver AR80201 pcs. - PDA5-01 measuring antenna - GSP-8101 Spectrum Analyzer - Sonata-AV Vibroacoustic Noise Reduction System 1 pcs. - Portable Noise Generator SELSP-21B 21 pcs. - Laboratory stand of the Stilpost video surveillance system 1 pc. - Laboratory stand of the video surveillance system "Synerget" 1 pc. - National Instruments hardware and software measuring complex 1 pc.
3	Laboratory Classes	Educational laboratory: laboratory in the field of Internet technologies	- SMART BOARD interactive whiteboard 1 pc. - Personal computer 9 pcs. - Laptop 4 pcs. - Router (LAN) 1 pc
4	Independent work	Library Reading Room	

VII. METHODOICAL INSTRUCTIONS FOR STUDENTS ON MASTERING THE DISCIPLINE

The educational process of teaching the discipline includes classroom classes (lectures, laboratory work) and independent work. Intermediate certification in the discipline - differentiated credit. The lecturer and teachers conducting classes control the attendance of all types of classroom classes.

Lectures are held with the obligatory use of presentations. The materials of lectures (presentations) are provided for the use of students by posting in the electronic space (electronic learning system lms.sfedu.ru).

Before conducting laboratory work, students must be familiarized with safety precautions when working in the laboratory. Before performing laboratory work, the teacher gives brief explanations on the purpose of the work and the process of its implementation. Based on the results of the work, students draw up a report on the implementation of laboratory work with the obligatory indication of the title of the work, the purpose of the work, the equipment used, the progress of work and the main results obtained, including the processing and analysis of the results of laboratory work. The defense of laboratory works is carried out, as a rule, at the next lesson in the form of an interview.

Independent work of students includes preparation for lectures, laboratory classes and tests.

Current and midterm control is carried out in classroom classes when defending reports on laboratory work and at tests. The maximum number of points for each type of control activities is indicated in the curriculum card of the discipline.

Students who, for a good reason, could not score the required number of points on the current and midterm control, can, in agreement with the teacher, eliminate debts before the end of the last week of theoretical training in the discipline.

VIII. FUND OF ASSESSMENT TOOLS

8.1. Passport of the Appraisal Tools Fund

Nop/ n	Competency Achievement Indicator Code	Name of the Valuation Tool
1	CPC-2.1	Test No1 (full-time) Exam questions and tickets (face-to-face)
2	CPC-2.3	Test No2 (full-time) Exam questions and tickets (face-to-face)
3	CPC-2.4	Laboratory work No1 (full-time) Laboratory Work No2 (full-time) Laboratory work No3 (full-time) Laboratory Work No4 (full-time) Laboratory work No5 (full-time) Laboratory work No6 (full-time) Laboratory Work No7 (full-time) Laboratory Work No8 (full-time)
4	CPC-5.1	Laboratory work No1 (full-time) Laboratory Work No2 (full-time) Laboratory work No3 (full-time) Laboratory Work No4 (full-time) Laboratory work No5 (full-time) Laboratory work No6 (full-time) Laboratory Work No7 (full-time) Laboratory Work No8 (full-time)
5	CPC-5.2	Laboratory work No1 (full-time) Laboratory Work No2 (full-time) Laboratory work No3 (full-time) Laboratory Work No4 (full-time) Laboratory work No5 (full-time) Laboratory work No6 (full-time) Laboratory Work No7 (full-time) Laboratory Work No8 (full-time)
6	PC-13.1. AI S -1.1	Laboratory work No1 (full-time) Laboratory Work No2 (full-time) Laboratory work No3 (full-time) Laboratory Work No4 (full-time) Laboratory work No5 (full-time) Laboratory work No6 (full-time) Laboratory Work No7 (full-time) Laboratory Work No8 (full-time)
7	PC-13.2 AI S -1.2	Laboratory work No1 (full-time) Laboratory Work No2 (full-time) Laboratory work No3 (full-time) Laboratory Work No4 (full-time) Laboratory work No5 (full-time) Laboratory work No6 (full-time) Laboratory Work No7 (full-time) Laboratory Work No8 (full-time)

8.2. Test No 1 (testing)

Type of assessment tool (subject of control): test work

List of supervised topics (modules): Module 1

Basic concepts and definitions. Threats and vulnerabilities of information systems. Data protection criteria and standards.

Goals and objectives of mastering the discipline. Introduction to Information Security. Basic concepts and definitions. Classification of sources of information security threats. Classification of threats to information security of an information system. Ensuring the security of the infrastructure of the information system. Methods for assessing threats to the security of an information system. Classification of intruders. Channels of penetration into the information system and channels of information leakage. Approaches to ensuring information security of information systems. Data protection criteria and standards.

Authentication methods and tools.

Basic concepts. Password authentication. One-time passwords. Functional authentication methods. Personal means of authentication. Biometric means of authentication. Authentication based on information associated with the subject. Multi-channel authentication.

Methods and means of authorization.

Access control. Security policies and models. Discretionary safety model. Mandatory security model. Role-based model of access control. A model of information flow security. Sandbox model. Thematic access model.

Technical protection of information.

Technical channels of information leakage. Use of technical channels of information leakage for technical intelligence. Classification of technical channels of information leakage. Technical channels of information leakage in the main technical means and systems. Electromagnetic leakage channels in the main technical means and systems. Electrical channels of information leakage in the main technical means and systems. Parametric channels of information leakage in the main technical means and systems. Technical channels of information leakage during data transmission. Mobile radio communication. Radio relay and space communication systems. Electrical channels of information leakage in wire communication lines. Interception of information from wire electrical communication lines. Interception of information from telephone communication lines. Interception of information from radiotelephone communication lines. Channels for information leakage from fiber-optic communication lines. Technical channels of speech information leakage. Speech information. Acoustic channels of speech information leakage. Vibroacoustic channels of speech information leakage. Acoustoelectric channels of speech information leakage. Acousto-fiber-optic channels for speech information leakage. Optoelectronic channels of speech information leakage. Parametric channels of speech information leakage. Technical channels of leakage of visual information. Covert surveillance of objects. Optical-mechanical devices. Thermal imagers and night vision devices. Covert photography of objects of observation. Covert photography of documents. Covert video surveillance. Material channels of information leakage. Integration of information leakage channels.

Organizational and legal support of information security.

Classification of information security methods. Classification and types of information resources. Restricted information. State secret. Confidential Information. Trade secret. Confidentiality stamps. Legal basis of the licensing and certification system. Licensing of activities related to state secrets and information protection. Certification of information security tools. Certification of activities for the technical protection of confidential information. Ethics of Artificial Intelligence. Legal regulation of artificial intelligence.

Artificial Intelligence in Information Security.

Problems of information security of artificial intelligence. Application of artificial intelligence in information security. Classification of information security systems using artificial intelligence. Classification of computer attacks on artificial intelligence. Methods of countering attacks on artificial intelligence. Products for information protection based on artificial intelligence. Directions for the development of artificial intelligence in information security. Federated learning. Differential confidentiality. Homomorphic encryption. Software products to protect the confidentiality and integrity of AI data. Protect your data from infection. Methods for recognizing anomalies.

Form of study: full-time

Bank of questions for the preparation of tests:

1. What is meant by information security and information protection?
2. What is the difference between the concepts of "information security" and "computer security"?
3. What are information resources, according to what principles are they classified?
4. Define the main components of information security?
5. What are the criteria for classifying threats to information security?
6. What is the intruder model?
7. What are the main causes, types and channels of information leakage?
8. What are the parts of the international standard ISO/IEC 15408 and its domestic counterpart GOST R ISO/IEC 15408 and what types of safety requirements does it contain?
9. What are the functional safety requirements in GOST R ISO/IEC 15408?
10. What are the safety trust requirements in GOST R ISO/IEC 15408?
11. What is the general scheme of identification/authentication procedures?
12. What are the methods of subject authentication?
13. What are the requirements for password subject authentication systems, their advantages and disadvantages?
14. How is subject authentication implemented using one-time passwords?
15. How is subject authentication implemented based on the "handshake" model?
16. What are the types of personal means of authentication of subjects, their main features, advantages and disadvantages?
17. Principles and features of the construction of static biometric means of authentication of subjects, their advantages, disadvantages, areas of application?
18. Principles and features of building dynamic biometric means of authentication of subjects, their advantages, disadvantages, areas of application?
19. How are errors of biometric means of authentication of subjects determined?
20. What is a security monitor and what are the requirements for it?
21. What is a security policy? In what form can it be presented?
22. What are the main types of security policies?
23. What is a security model. What is the general scheme of a security model?
24. How is the discretionary security model implemented?
25. How is the mandate security model implemented?
26. How is the role model of security implemented?
27. What technical channels of information leakage arise when information is processed by means of computer technology?
28. What are the portable means of technical reconnaissance?
29. What methods and means of protection against technical intelligence are used?
30. What are the methods and equipment for protecting telephone lines?
31. What are the means of protection against unauthorized access to acoustic information?
32. What are the means of detecting radio stashes?
33. What are the legal and organizational methods of information protection?

34. How are information resources classified by access categories?
35. The concept of state secrets and the principles of their protection?
36. What are the basic principles of trade secret protection?
37. What are the basic principles of protection of professional and official secrets?
38. What is the legal basis of the licensing and certification system?
39. How is the licensing of activities for the protection of state secrets carried out?
40. What are the main principles of certification of information security tools?
41. How is licensing and certification carried out in the field of confidential information protection?
42. List the principles of legal regulation of artificial intelligence.
43. What is the ethics of artificial intelligence?
44. Indicate the risks of using artificial intelligence.
45. What methods are used to detect and mitigate risks?
46. List the problems of information security of artificial intelligence.
47. Why and how is artificial intelligence used in information security?
48. Provide a classification of information security systems using artificial intelligence.
49. Provide a classification of computer attacks on artificial intelligence.
50. List methods to counter attacks on artificial intelligence.
51. What products are used to protect information based on artificial intelligence?
52. Tell us about the directions of development of artificial intelligence in information security.
53. What is federated learning?
54. Talk about differential privacy.
55. What is homomorphic encryption?
56. List software products to protect the confidentiality and integrity of AI data.
57. Tell us about data protection from infection.
68. List the methods for recognizing anomalies.

The test allows you to assess the level of formed knowledge, skills and abilities within the framework of the competencies formed by these sections of the discipline, including in the course of independent work of the student.

The test is carried out in writing. When writing an answer, the student needs to answer 4 control questions. Each of the 4 questions of the test is selected (randomly) from the bank of questions of the corresponding section: the first question is from the bank of questions on section 1, the second question is from the bank of questions on section 2, etc.

An example of the content of the ticket for the test No1:

**Test No1
on the discipline "Security of Information Technologies"**

1. What is meant by information security and information protection?
2. What is the general scheme of identification/authentication procedures?
3. What technical channels of information leakage arise when information is processed by means of computer technology?
4. The place of information law in the general system of law?

8.3. Test No 2 (testing)

Type of assessment tool (subject of control): test work

List of controlled topics (modules): Cryptographic methods of information security and cryptanalysis. Symmetric and asymmetric algorithms (AES, GOST, Kuznechik, RSA). Hashing, EDS, PKI. AI threats: faster key guessing, ciphertext analysis. Requirements for durability and certification.

Cryptographic methods of information protection and cryptanalysis.

Typical implementation errors: weak RNGs, retry IV, poor key management. Methods of cryptanalysis. The role of AI in automating attacks on cryptosystems. FSB requirements for domestic algorithms. Post-quantum cryptography.

Cryptographic methods of information protection and cryptanalysis.

TLS, IPsec, PGP protocols. Security in mobile and IoT devices. Use AI to monitor PKI integrity and detect anomalies in cryptosystems. Incident analysis: Heartbleed, Logjam. Updating and maintaining cryptographic libraries.

Malicious software.

Classification: viruses, worms, trojans, rootkits, ransomware. Polymorphism and metamorphism. Use of AI by attackers: phishing generation, deepfake, EDR bypass. The use of AI in next-generation antivirus systems.

Malicious software.

Targeted attacks (APTs), multi-stage chains of compromise. AI for analyzing the behavior of software in the sandbox. Examples: Stuxnet, WannaCry, NotPetya. Countermeasures: backup, patching, cyber hygiene.

Security of operating systems.

Security architecture: privilege levels, process isolation. Mechanisms: SELinux, AppArmor, UAC, SIP. Exploits: buffer overflow, race conditions. Using AI to monitor system calls and detect rootkits.

Security of operating systems.

Kernel protection, TPM, Secure Boot. Virtualization and containerization as isolation tools. AI to automatically configure security policies and analyze OS integrity. Evaluation standards: Common Criteria, EAL.

Protection of information in computer networks.

Network attacks: MITM, ARP-spoofing, DDoS. Protection tools: firewall, IDS/IPS, segmentation. Protocols: IPsec, TLS, WPA3. AI for encrypted traffic analysis, anomaly detection, and automatic network segmentation.

Protection of information in computer networks.

Zero Trust architectures, SDP, secure remote access. AI in cloud and hybrid information security. Predict threats and automate responses. Standards: NIST CSF, CIS Controls.

Form of study: full-time

Bank of questions for the preparation of tests:

1. Attacks on cryptographic systems and algorithms: explain the concepts of cryptoattack and cryptographic strength; information available to the attacker during the analysis of the cipher; the main indicators of cryptographic strength; Give and explain the main methods of cryptanalysis of ciphers.
2. Stream ciphers: principles of encryption and decryption of information when using gamma (key pseudorandom sequence), basic requirements for pseudorandom sequences).

3. Block ciphers: principles of encryption and decryption of information; modes of operation of block ciphers; the capabilities that the attacker has when analyzing the block cipher; Basic requirements for a block cipher.
4. Russian block encryption standard "Grasshopper": basic parameters, structure of the encryption round (cycle) and principle of operation, the procedure for deploying the initial key into round (working) connections, modes of use.
5. Asymmetric cryptographic systems: causes of appearance and main ideas; unilateral functions; factorization; discrete logarithm and discrete root.
6. RSA cryptographic algorithm: the mathematical problem behind the algorithm; the principle of generating the user's public and private keys; the principle of message encryption; Give an example of message encryption.
7. El-Gamal's cryptographic system: a mathematical problem underlying the algorithm; the principle of generating the user's public and private keys; The principle of message encryption.
8. Hash functions: the purpose of hash functions; basic requirements for hash functions; provide a brief description of the SHA-3 hash function.
9. Hash functions: the purpose of hash functions; basic requirements for hash functions; give a brief description of the hashing function GOST R 34.11-2012 (Streebog).
10. Electronic signature: appointment; major violations of the authenticity of messages; electronic signature based on the RSA algorithm.
11. Key Management: Assignment; Explain components such as key generation, key accumulation, and key distribution. provide a brief description of the Diffie-Hellman public key distribution scheme.
12. Authentication protocols with symmetric encryption algorithms (What is the idea behind strong authentication? Describe and explain the following methods: one-way authentication based on timestamps; one-way authentication based on random numbers; two-way authentication using random values).
13. What is malware, types, goals and objectives of its creation and use?
14. What are the methods of protection against malware?
15. What OS defects can lead to the creation of information leakage channels? Provide and explain the main vulnerabilities of the software.
16. Explain the method of detecting malware based on signature search, give its advantages and disadvantages.
17. Explain the method of detecting malware based on heuristic search, give its advantages and disadvantages.
18. Explain the method of detecting malware based on monitoring of information flows, give its advantages and disadvantages.
19. Explain the method of detecting malware based on integrity control, give its advantages and disadvantages.
20. Explain the method of detecting malware based on software traps, give its advantages and disadvantages.
21. The concept of a secure operating system (Typical attacks on operating systems. What is a secure operating system, security policy and adequate security policy?
22. Tasks of hardware for information protection in operating systems. (What are the main approaches to ensuring the protection of RAM? What is the purpose of scheduling tasks in a multitasking OS? How are RAM segments protected in x86 and x86-64 processors?)
23. Operating System Audit Subsystem. (Overview of auditing: Why should audit functions be included in the operating system? Requirements for the audit subsystem.)

24. Encrypt files in Windows when using Encrypted File System (EFS) (What are the main reasons for the unreliability of cryptographic systems, give a brief explanation of these reasons).
25. Encryption in popular archivers. (Give the main reasons for the unreliability of cryptographic systems when used in archivers, give a brief explanation of these reasons. How is encryption carried out in popular archivers? What is the strength of these versions of archivers?)
26. Software tools for transparent disk encryption (What are the possibilities of software tools for transparent disk encryption?)
27. Provide a brief description and characteristics of the AES-NI and NX-Bit (ND-Bit) instruction extensions.
28. Access control systems based on virtualization technology. (Give the main types of virtualization technologies, their advantages and disadvantages.)
29. ASLR Address Space Location Randomization Technology. (Purpose, ways to bypass protection, advantages and disadvantages).
30. Basic functions of information protection in data networks (explain the following components: confidentiality, authentication, integrity, non-deception, access control, availability of resources).
31. Firewall functions (Where should firewalls be installed? Main tasks solved by firewalls. What are the characteristics of firewalls? Explain the following functions of firewalls – traffic filtering and intermediation functions).
32. Shielding router (purpose, principle of operation, advantages and disadvantages).
33. Session-level gateway (purpose, principle of operation, advantages and disadvantages).
34. Application layer gateway (purpose, principle of operation, advantages and disadvantages).
35. Criteria for assessing the quality of firewalls (What groups of requirements should firewalls meet? Give the main security classes of firewalls in accordance with the guidelines of the FSTEC of Russia. What requirements should firewalls meet in accordance with security classes?)
36. Virtual secure VPN networks. (What is a virtual secure network? The principle of the tunneling mechanism, the criteria for the security of transmitted data and how to ensure them.)
37. IPSec network layer protection protocol (purpose, features of VPN tools of this level, cryptographic technologies used, main components, advantages and disadvantages).
38. SSL/TLS session layer protection protocols (purpose, features of VPN facilities of this level, cryptographic technologies used, main stages of formation and maintenance of a protected connection, advantages and disadvantages).
39. L2F and L2TP channel layer protection protocols (purpose, features of VPN facilities of this level, cryptographic technologies used, main stages of formation and maintenance of a protected connection, advantages and disadvantages).
40. Adaptive network security management (prerequisites for use; explain the following main components – risk assessment, security analysis, attack detection; explain the purpose and principle of operation of security scanners).
41. Attack detection systems (purpose of attack detection systems; provide and explain methods for analyzing network information – statistical method, expert systems, neural networks; advantages and disadvantages of these methods).

An example of the content of the ticket for the test work in module 2:

Test No2 on the discipline "Security of Information Technologies"
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1. Stream ciphers: principles of encryption and decryption of information when using gamma (key pseudorandom sequence), basic requirements for pseudorandom sequences).
2. What is malware, varieties, goals and objectives of its creation and use?
3. The concept of a secure operating system (Typical attacks on operating systems. What is a secure operating system, security policy and adequate security policy?)
4. Firewall functions (Where should firewalls be installed? Main tasks solved by firewalls).

Evaluation criteria:

The test is evaluated with a maximum number of points equal to 10, which corresponds to the correct answers to all questions. The correct answer to each of the 4 questions is evaluated at a maximum of 2.5 points, the value of the score may decrease in proportion to the completeness of the answer or the number of correct answers to the sub-questions. After summing up the points for all correct answers, the sum of points scored is rounded to an integer value.

8.4. Laboratory Work No 1

Type of assessment tool (subject of control): Preparation and submission of a report on laboratory work

List of monitored topics (modules): Analysis of spurious electromagnetic emissions using AI for anomaly detection

Using machine learning to classify EMR and identify hidden transmitters based on spectrograms.

Form of study: full-time

Topic: Analysis of spurious electromagnetic emissions using AI for anomaly detection

Purpose of the work

Learn how to collect radio frequency spectrum using software-defined radio (SDR), identify signs of unauthorized transmission, and apply machine learning techniques to automatically detect anomalies on the air.

Task:

Preparation of equipment and software

Connect the HackRF One SDR receiver to your computer.

Install the software:

GNU Radio Companion or CubicSDR for spectrum visualization;

Python 3.x with libraries: numpy, scipy, matplotlib, scikit-learn (or TensorFlow/PyTorch if you have experience).

Make sure that the HackRF drivers are installed (hackrf_info in the terminal should display the device).

Ether Data Collection

Select the frequency range typical for spurious emissions of digital equipment (for example, 30-500 MHz).

Record an I/Q stream (complex float32) for 5 to 10 minutes under "normal" conditions (no active transmitters).

Repeat the recording under the simulated threat conditions:

Place a compact Wi-Fi/Bluetooth transmitter or USB modem near your PC.

Enable data transfer (e.g., file download).

Save both datasets in .sigmf-data or .bin format.

Pre-Processing and Visualization

Convert I/Q data into spectrograms (time $\times$ frequency $\times$ power).

Visually compare "normal" and "abnormal" airwaves: note the emergence of new narrowband or broadband signals.

Applying AI to Detect Anomalies

Form a dataset of spectrograms:

Class 0 — "normal";

Class 1 — "Anomaly" (the presence of a hidden transmitter).

Train a simple machine learning model (such as Random Forest, SVM, or CNN) for binary classification.

Evaluate the accuracy of the model (accuracy, precision, recall).

(Optional): Try using unsupervised methods (e.g., Isolation Forest or Autoencoder) to detect anomalies without labeled data.

Analysis of results and conclusions

Answer the questions:

What are the most informative signs for detecting hidden transmitters?

How resistant is the model to noise and changes in the air?

What are the limitations of the AI-based approach in technical information security?

Report Requirements

The report must contain:

a brief description of the hardware and software complex;

screenshots of the "normal" and "anomalies" spectrograms;

a description of the architecture of the AI model and its quality metrics;

conclusions on the applicability of AI for monitoring technical leakage channels;

recommendations for improving the detection system.

Safety measures

Do not transmit signals without permission (HackRF One is safe in receive mode).

Work within a dedicated training frequency band.

Do not record signals that contain personal data or protected information.

Connect the output of the instrument to your computer's sound card or audio interface.

Simulation of information leakage

In the room, play a speech signal (for example, the announcer's reading of the text).

Record the vibration signal from the object surface using the software provided with the Piranha (or via Audacity / Python).

Repeat the recording in active protection conditions:

turn on the masking noise generator (for example, white noise or speech-like signal);

or apply a mechanical damping pad to the surface.

Save both audio files in .wav format (16-bit, 44.1 kHz).

Signal Analysis

Visualize the time and frequency characteristics (spectrograms) of the original and protected signals.

Evaluate speech intelligibility in a vibration signal "before" and "after" protection (subjectively and with automatic transcription, e.g. via Whisper API or Vosk).

AI filter development

Form a dataset from the "polluted signal → pure noise/masking" pairs.

Train a neural network noise reduction model (e.g., based on U-Net, LSTM, or simple CNN autoencoder architecture) to remove speech components from the vibration signal.

(Resource-constrained alternative): Apply a pre-trained model (e.g., noisereduce + scikit-learn Isolation Forest) to identify and suppress anomalous (speech) fragments.

Assessment of the effectiveness of protection

Compare the quality of speech suppression before and after the AI filter:

by metrics (SNR, PESQ);

based on the results of automatic transcription (WER - Word Error Rate).

Draw a conclusion about the applicability of AI approaches in systems for technical protection of information from vibroacoustic leaks.

Report Requirements

The report must contain:

Pilot plant diagram with ST-500 parameters

screenshots of spectrograms of the original and processed signals;

description of the architecture of the AI model and its metrics;

comparative analysis of the effectiveness of traditional and AI protection methods;

conclusions and recommendations for the integration of AI into means of countering technical leakage channels.

Safety and Ethics Measures

The work is carried out only in a training laboratory with permitted sound sources.

It is forbidden to record real conversations or confidential information.

All data is deleted when the work is completed.

5. What are functional safety requirements in GOST R ISO/IEC 15408?
6. What are the security trust requirements in GOST R ISO/IEC 15408?
7. What is the general scheme of identification/authentication procedures?
8. What are the ways to authenticate subjects?
9. What are the requirements for subject password authentication systems, their advantages and disadvantages?
10. How is subject authentication implemented with one-time passwords?
11. How is subject authentication implemented based on the "handshake" model?
12. What are the types of personal means of authentication of subjects, their main features, advantages and disadvantages?
13. Principles and features of building static biometric means of authentication of subjects, their advantages, disadvantages, areas of application?
14. Principles and features of building dynamic biometric means of authentication of subjects, their advantages, disadvantages, areas of application?
15. How are biometric subject authentication errors determined?
16. What is a security monitor and what are the requirements for it?
17. What is a security policy? In what form can it be presented?
18. What are the main types of security policies?
19. What is a security model? What is the general outline of the security model?
20. How is the discretionary security model implemented?
21. How is the mandatory security model implemented?
22. How is the role-based security model implemented?
23. What technical channels of information leakage arise when information is processed by means of computer technology?
24. What are the types of portable technical reconnaissance tools?
25. What methods and means of protection against technical intelligence are used?
26. What are the methods and equipment for protecting telephone lines?
27. What are the means of protection against DDS to acoustic information?
28. What are the means of detecting radio bookmarks?
29. What are the legal and organizational methods of information protection?
30. How are information resources classified by access categories?
31. The concept of state secrets and the principles of its protection?
32. What are the basic principles of trade secret protection?
33. What are the basic principles for the protection of professional and official secrets?
34. What is the legal basis for the licensing and certification system?
35. How is the licensing of activities for the protection of state secrets carried out?
36. What are the main principles of certification of information security tools?
37. How is the privacy protection licensing and certification processed?
38. Principles of legal regulation of artificial intelligence.
39. Ethics of Artificial Intelligence.
40. Problems of information security of artificial intelligence.
41. Application of artificial intelligence in information security.
42. Classification of information security systems using artificial intelligence.
43. Classification of computer attacks on artificial intelligence.
44. Methods of countering attacks on artificial intelligence.

45. Products for information protection based on artificial intelligence.
46. Directions for the development of artificial intelligence in information security.
47. Federated learning.
48. Differential confidentiality.
49. Homomorphic encryption.
50. Software products to protect the confidentiality and integrity of AI data.

Module 2:

1. The Russian block encryption standard "Grasshopper": basic parameters, structure of the encryption round, key unfolding, modes of use. Are attacks on the Grasshopper using machine learning possible? Justify.
2. RSA algorithm: mathematical basis, key generation, encryption. How can AI speed up factorization or key guessing? What countermeasures are in place?
3. El-Gamal's cryptosystem: mathematical problem, key generation, encryption. Can AI be used to analyze discrete logarithms in the context of this system?
4. Hash functions: purpose, requirements, description of SHA-3. How is AI used to find collisions or generate images?
5. Hash function of GOST R 34.11-2012 (Streebog): features, safety. Compare its resistance to classic and AI-accelerated attacks.
6. Electronic signature: assignment, authentication violations, RSA-based digital signature. How can AI be used to forge signatures or detect forgeries?
7. Key management: generation, distribution, accumulation. The Diffie–Hellman protocol. How can AI help detect anomalies in key distribution (e.g., in PKI)?
8. Authentication protocols with symmetric algorithms: strong authentication, timestamp and random number methods. How is AI used in Adaptive Authentication?
9. Malware detection methods: a) signature search – advantages and disadvantages; (b) heuristic analysis; c) Machine learning and behavioral analysis (ML/UEBA) – principles, benefits, risks (e.g., model traversal with adversarial examples).
10. Monitoring of information flows and integrity control as methods of protection against malware. How does AI reinforce these approaches?
11. Software traps (honeypots/honeynets): purpose, application. How is AI used to analyze the behavior of attackers in traps?
12. OS audit subsystem: purpose, requirements. How is AI used to automatically analyze logs and identify insider threats?
13. OS hardware: AES-NI, NX/DEP, ASLR. How can AI be used to circumvent these mechanisms (e.g., through training on memory leaks)?
14. Virtualization and isolation: types, role in security. How does AI help detect hypervisor or container compromise?
15. ASLR Technology: Purpose, Bypass, Protect. Can AI predict memory addresses based on program behavior?
16. Firewalls and gateways: shielding router, session and application layer gateways. How is AI used in the next-generation firewall (NGFW) for traffic analysis?
17. IPSec, SSL/TLS, L2TP/L2F protocols: destination, cryptography, connection establishment stages. How is AI used to analyze encrypted traffic without decryption?
18. Adaptive network security management: risk assessment, vulnerability scanners. How does AI automate risk assessment and vulnerability prioritization?
19. Attack detection systems (IDS/IPS): a) statistical methods; (b) Expert systems; c) neural networks and deep learning – principles, examples of architectures (CNN, LSTM), advantages and disadvantages (interpretability, resistance to adversarial attacks).
20. Trusted AI: definition, key properties (robustness, explainability, functional safety). What threats exist for AI systems in information security?
21. AI risk management: stages, assessment methods, standards. How can AI risk management be integrated into an organization's overall information security policy?

22. Generative models (LLM, deepfake) in cyberattacks: phishing, voice/video spoofing, social engineering automation. What technical and organizational countermeasures exist?
23. Zero Trust Architecture: principles, components. How is AI used to dynamically verify subjects and make access decisions?
24. Protection of AI systems from malicious influences: adversarial attacks, data poisoning, model stealing. Give examples and methods of protection.
25. Ethical and legal aspects of the use of AI in information security: responsibility for AI solutions, the requirements of FZ-152 and FZ-187 for data processing with the participation of AI, international experience (AI Act).

Examination evaluation criteria:

The minimum number of points required for admission to the exam is 38 points. The exam ticket contains 2 questions, the first question is selected from the list of questions in Module 1, the second question is selected from the list of questions in Module 2.

40-34 points are given to a student if:

- answers to all questions of the exam ticket are given in full;
- Answers are characterized by an understanding of the principles, coherence, logic, sufficient volume;
- single errors are allowed, which are corrected in the process of answering by the examinee himself.

33-28 points are given to a student if:

- answers to all tasks of the exam ticket are given in full and are characterized by an understanding of the principles, answers to additional questions are characterized by coherence, logic, but minor errors or inaccuracies are allowed;
- some of the tasks of the exam ticket were completed with minor shortcomings, the answers to additional questions were given in full;
- single errors that were corrected by the examiners in the process of answering are allowed.

27-22 points are given to a student if:

- one or two tasks of the exam ticket were completed with significant errors;
- Answers to additional questions are characterized by insufficient coherence, logic, misunderstanding of the schemes;
- when answering, the student makes mistakes that he is not able to correct on his own.

0 points are given to a student if:

- tasks are completed fragmentarily with significant errors;
- Answers are characterized by a lack of coherence, logic, lack of understanding of the schemes;
- when answering, the student makes gross mistakes that he is not able to correct on his own.

The final score depends on the sum of points for all types of control and is given according to the following scale:

less than 60 points – unsatisfactory,

60-70 points – satisfactory;

71-84 points – good;

85-100 points – excellent